

Trip Generation Analysis - SVC790 Part A

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1 Overview

Dependent variable: `N_trips`. Predictors: `Nr_People_H`, `Elderly`, `Educ_HS`, `Educ_Tertiary`, `Workers_Fulltime`, `Workers_Parttime`, `Learners_students`, `Driv_Lic`.

Every task below is done two ways:

- **Version A - Base R:** uses only functions shipped with base R / the `stats` package (no extra packages, beyond what is unavoidably needed to read the `.xlsx` file in the first place).
- **Version B - Packages:** the same task using a common contributed package that makes it faster or nicer to look at.

Both versions compute the same underlying numbers; only the mechanics differ.

```
# readxl has no base-R equivalent (base R cannot read .xlsx files), so it is  
# loaded once here as shared infrastructure rather than as part of a  
# "package version" of any one task.  
ensure_pkg("readxl")  
  
if (!dir.exists("figures")) dir.create("figures")
```

1.1 Load data

The sheet has a title block in rows 1-9; the real header row is row 10, so we skip the first 9 rows when reading.

```
xlsx_path <- "RLubega SVC790_Part A_2026.xlsx"  
  
raw <- read_excel(xlsx_path, sheet = "Table 1", skip = 9)  
  
predictors <- c("Nr_People_H", "Elderly", "Educ_HS", "Educ_Tertiary",  
               "Workers_Fulltime", "Workers_Parttime",  
               "Learners_students", "Driv_Lic")  
  
vars <- c("N_trips", predictors)  
  
data <- raw[, vars]  
data[] <- lapply(data, function(x) as.numeric(x))  
  
cat("Rows read:", nrow(data), "\n")  
  
## Rows read: 5387  
  
cat("Rows with at least one NA among analysis variables:",  
    sum(!complete.cases(data)), "\n")
```

```
## Rows with at least one NA among analysis variables: 71
```

2 1. Correlation matrix and plot

The correlation values themselves come from base R `cor()` either way; only the plot differs between the two versions.

```
cor_mat <- cor(data[, vars], use = "complete.obs")
round(cor_mat, 3)
```

```
##           N_trips Nr_People_H Elderly Educ_HS Educ_Tertiary
## N_trips      1.000      0.411   0.196   0.240      0.121
## Nr_People_H   0.411      1.000   0.568   0.139      0.026
## Elderly       0.196      0.568   1.000   0.043      0.018
## Educ_HS       0.240      0.139   0.043   1.000     -0.279
## Educ_Tertiary 0.121      0.026   0.018  -0.279      1.000
## Workers_Fulltime 0.230      0.118   0.069   0.252      0.343
## Workers_Parttime 0.142      0.017   0.005   0.044     -0.028
## Learners_students 0.545      0.271   0.135  -0.013     -0.010
## Driv_Lic      0.108      0.017  -0.013   0.074      0.385
## Workers_Fulltime Workers_Parttime Learners_students Driv_Lic
## N_trips 0.230 0.142 0.545 0.108
## Nr_People_H 0.118 0.017 0.271 0.017
## Elderly 0.069 0.005 0.135 -0.013
## Educ_HS 0.252 0.044 -0.013 0.074
## Educ_Tertiary 0.343 -0.028 -0.010 0.385
## Workers_Fulltime 1.000 -0.164 -0.012 0.373
## Workers_Parttime -0.164 1.000 0.025 -0.041
## Learners_students -0.012 0.025 1.000 -0.073
## Driv_Lic 0.373 -0.041 -0.073 1.000
```

2.1 Version A - Base R: correlation heatmap (`image()`)

Base R has no built-in ellipse-style correlation plot, so the base-R equivalent here is a colour-coded heatmap built with `image()`, with the correlation values overlaid as text.

```
plot_cor_heatmap <- function(m, title = "") {
  n <- ncol(m)
  m_flipped <- m[n:1, ]
  op <- par(mar = c(1, 7, 3, 1))
  image(1:n, 1:n, t(m_flipped), axes = FALSE, xlab = "", ylab = "",
        col = colorRampPalette(c("firebrick3", "white", "steelblue4"))(101),
        zlim = c(-1, 1), main = title)
  axis(3, at = 1:n, labels = colnames(m), las = 2, cex.axis = 0.7)
  axis(2, at = 1:n, labels = rev(rownames(m)), las = 2, cex.axis = 0.7)
  for (i in 1:n) {
    for (j in 1:n) text(j, n - i + 1, sprintf("%.2f", m[i, j]), cex = 0.55)
  }
}
```

```

box()
par(op)
}

plot_cor_heatmap(cor_mat, "Correlation matrix: N_trips and 8 predictors")

```

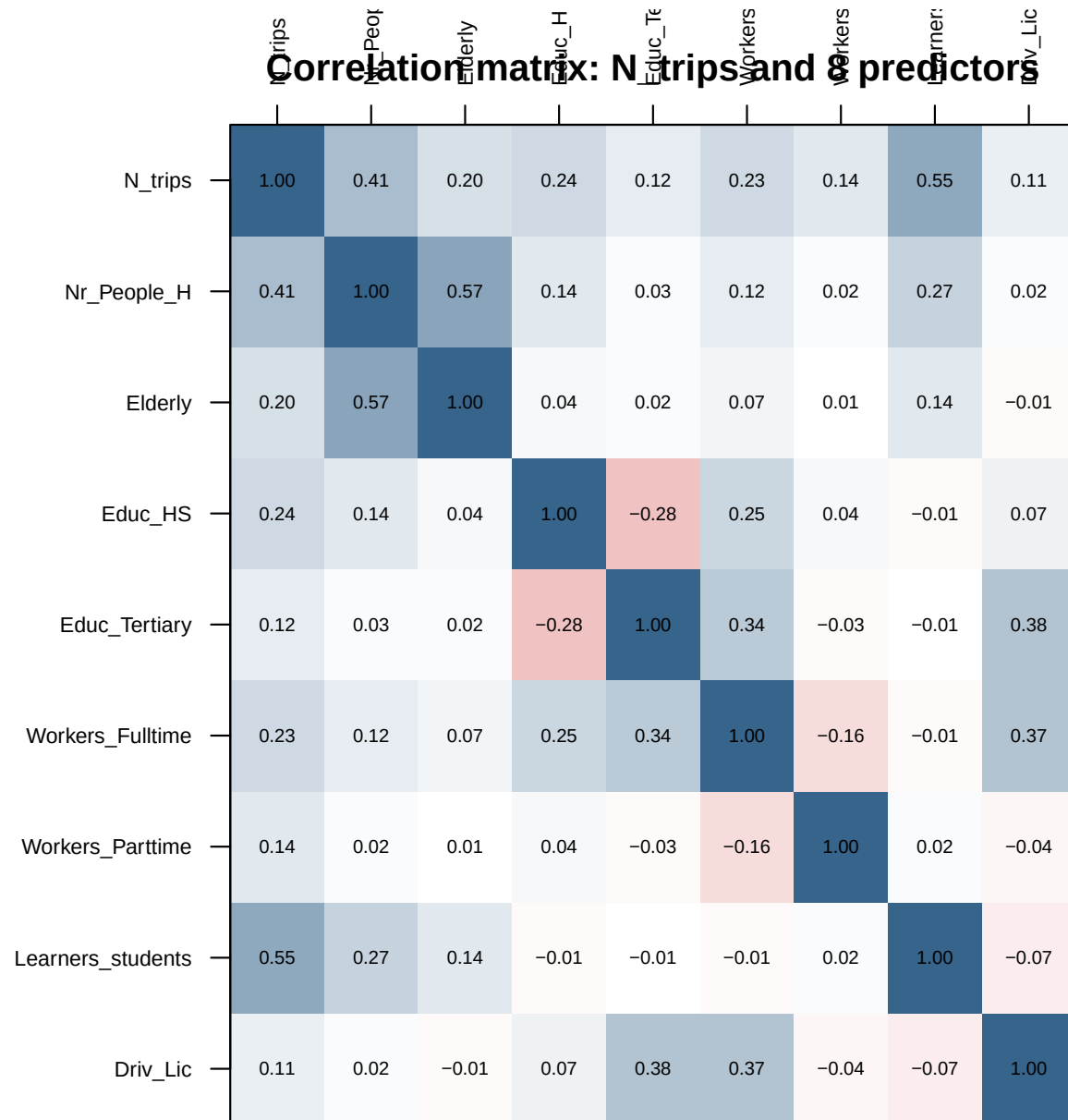


Figure 1: Base R correlation heatmap

2.2 Version B - Package (corrplot): ellipse plot

```
ensure_pkg("corrplot")

corrplot(cor_mat, method = "ellipse", type = "full",
         tl.col = "black", tl.srt = 45,
         addCoef.col = "black", number.cex = 0.6,
         title = "Correlation matrix: N_trips and 8 predictors",
         mar = c(0, 0, 2, 0))
```

3 2. Scatterplots of N_trips against each predictor

3.1 Version A - Base R

```
plot_scatter_grid <- function() {
  par(mfrow = c(2, 4), mar = c(4, 4, 2, 1))
  for (p in predictors) {
    plot(data[[p]], data[["N_trips"]],
         xlab = p, ylab = "N_trips",
         main = paste("N_trips vs", p),
         pch = 16, col = rgb(0, 0, 1, 0.3))
    abline(lm(data[["N_trips"]] ~ data[[p]]), col = "red", lwd = 2)
  }
}
plot_scatter_grid()
```

3.2 Version B - Package (ggplot2)

```
ensure_pkg("ggplot2")
ensure_pkg("tidyr")

data_long <- tidyr::pivot_longer(data, cols = all_of(predictors),
                                names_to = "predictor", values_to = "value")

gg_scatter <- ggplot2::ggplot(data_long, ggplot2::aes(x = value, y = N_trips)) +
  ggplot2::geom_point(alpha = 0.25, colour = "steelblue4") +
  ggplot2::geom_smooth(method = "lm", formula = y ~ x, colour = "firebrick3",
                      se = FALSE, linewidth = 0.8) +
  ggplot2::facet_wrap(~predictor, scales = "free_x", ncol = 4) +
  ggplot2::labs(x = NULL, y = "N_trips",
               title = "N_trips vs each of the 8 predictors") +
```

Correlation matrix: N_trips and 8 predictors

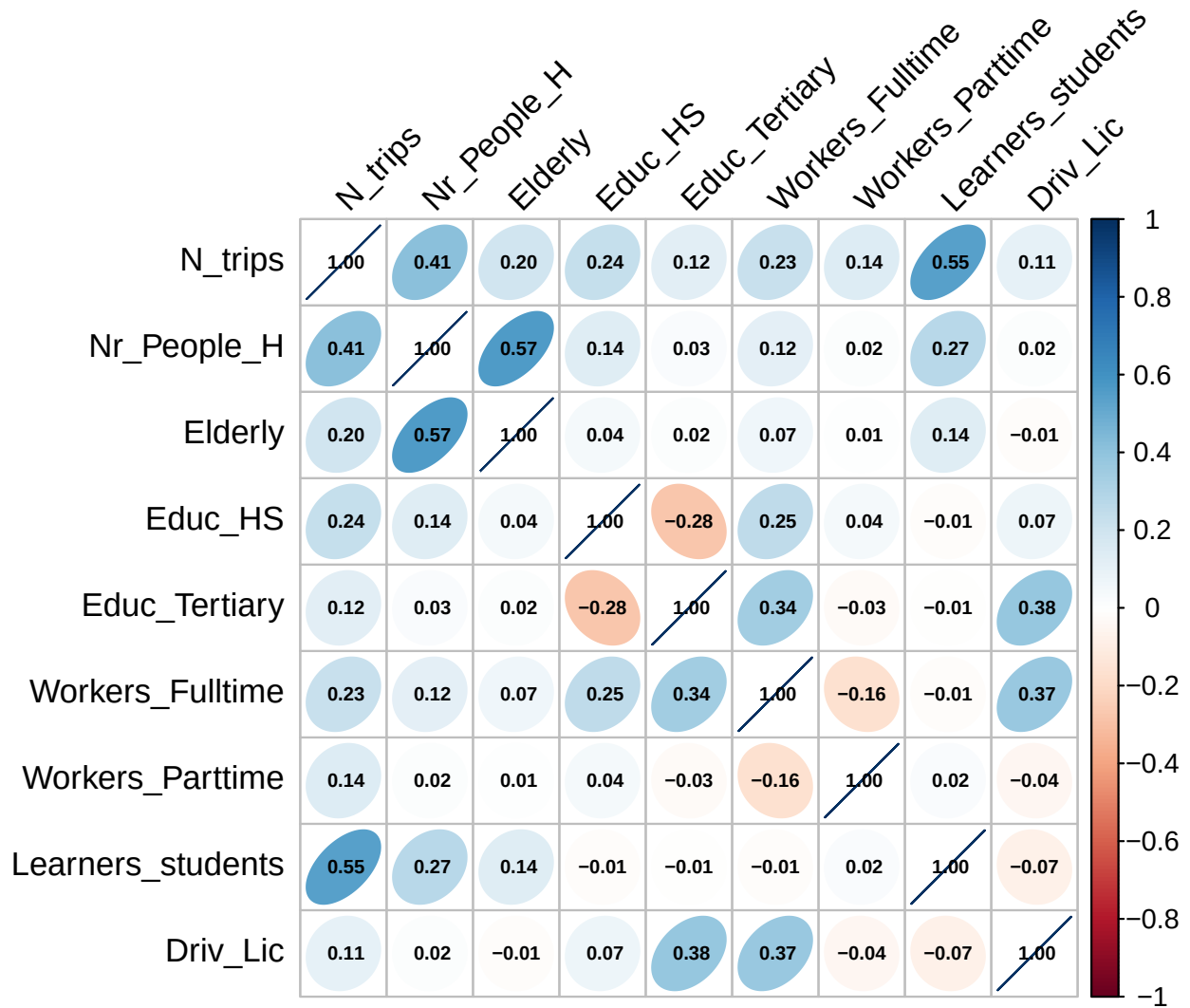


Figure 2: Correlation matrix: N_trips and 8 predictors (corrplot ellipse)

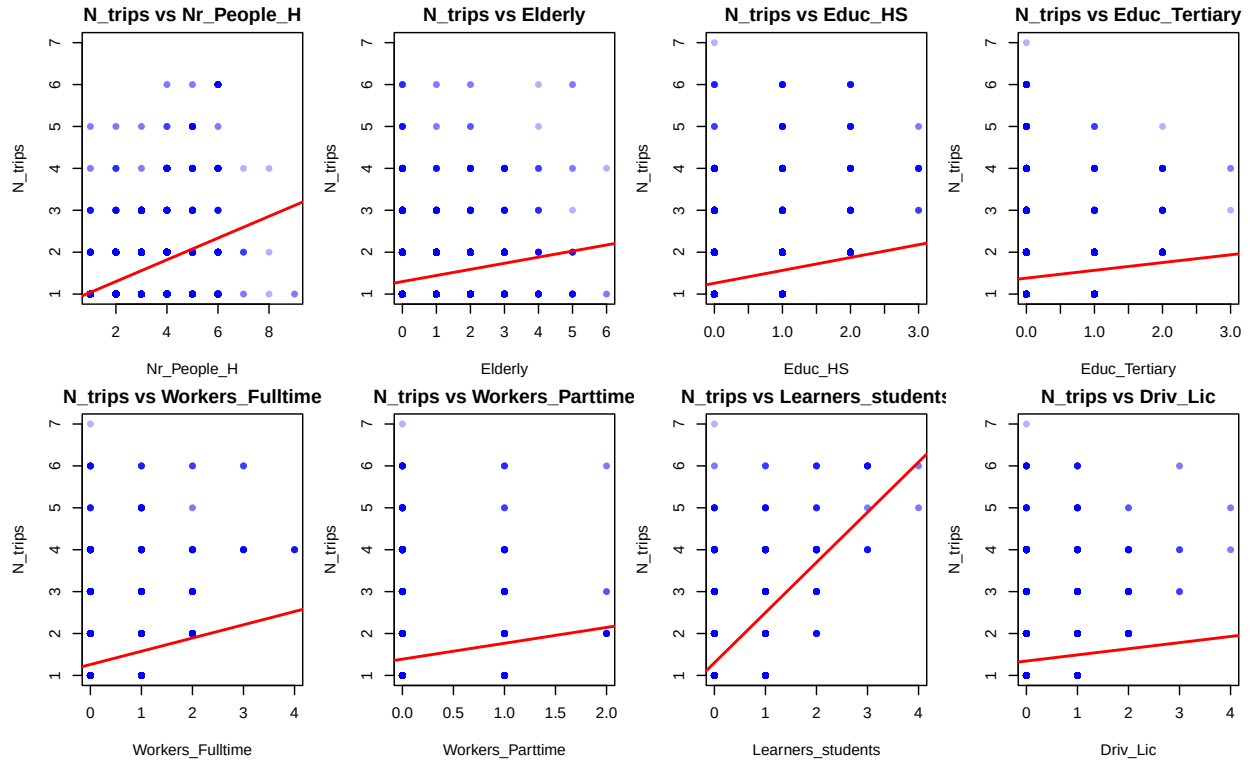


Figure 3: N_trips vs each of the 8 predictors (base R)

```
ggplot2::theme_bw(base_size = 9)

gg_scatter
```

4 3. Variance Inflation Factors (VIF)

4.1 Version A - Base R (manual auxiliary regressions)

Each predictor is regressed on the other 7 using base R `lm()`, and

$$VIF = \frac{1}{1 - R_{aux}^2}.$$

No car package is used for this version.

```
vif_data <- data[complete.cases(data[, predictors]), predictors]

r2_aux <- sapply(predictors, function(p) {
  other <- setdiff(predictors, p)
  form <- as.formula(paste(p, "~", paste(other, collapse = " + ")))
  summary(lm(form, data = vif_data))$r.squared
```

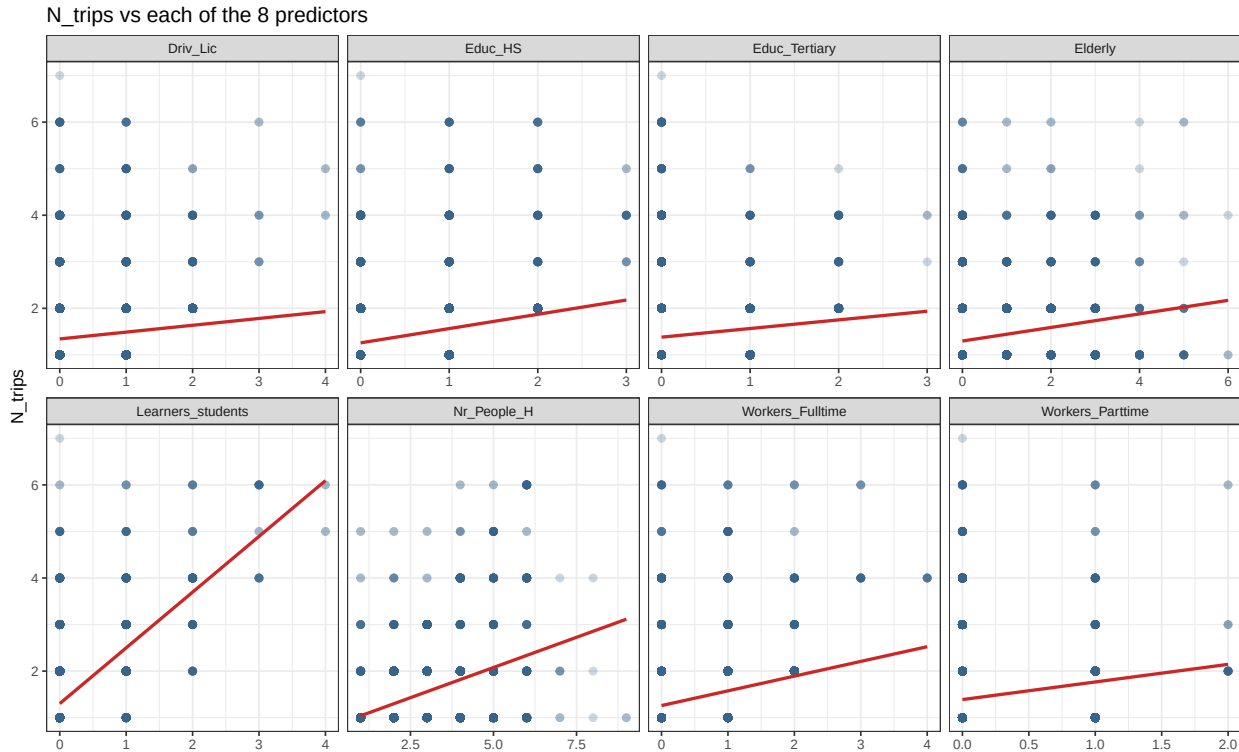


Figure 4: N_trips vs each of the 8 predictors (ggplot2)

```

})

vif_table_baseR <- data.frame(Predictor = predictors,
                              R2_aux = round(r2_aux, 4),
                              VIF = round(1 / (1 - r2_aux), 3))

vif_table_baseR_display <- vif_table_baseR
if (knitr::is_latex_output()) {
  vif_table_baseR_display$Predictor <- gsub("_", "\\_", vif_table_baseR_display$Predictor)
}
colnames(vif_table_baseR_display) <- c("Predictor", "$R^2_{aux}$", "VIF")

kable_safe(vif_table_baseR_display, row.names = FALSE, escape = FALSE,
           caption = "Base R manual VIF")

```

4.2 Version B - Package (car::vif())

car::vif() reads the VIFs off a fitted model object directly.

```
ensure_pkg("car")
```


Table 1: Base R manual VIF

Predictor	R^2_{aux}	VIF
Nr_People_H	0.3795	1.612
Elderly	0.3253	1.482
Educ_HS	0.2473	1.329
Educ_Tertiary	0.3363	1.507
Workers_Fulltime	0.3151	1.460
Workers_Parttime	0.0407	1.042
Learners_students	0.0829	1.090
Driv_Lic	0.2286	1.296

Table 2: car::vif() result

Predictor	VIF
Nr_People_H	1.612
Elderly	1.482
Educ_HS	1.329
Educ_Tertiary	1.507
Workers_Fulltime	1.460
Workers_Parttime	1.042
Learners_students	1.090
Driv_Lic	1.296

```

full_formula <- reformulate(predictors, response = "N_trips")
vif_full_model <- lm(full_formula, data = data)

vif_table_car <- data.frame(Predictor = names(car::vif(vif_full_model)),
                           VIF = round(car::vif(vif_full_model), 3))

kable_safe(vif_table_car, row.names = FALSE,
           caption = "car::vif() result")

```

Both versions agree (checked with `all.equal()` above): manual VIFs are all below 2, so multicollinearity is not severe among the 8 predictors.

5 4. Fit the six models

`lm()` itself is base R either way; what differs between the two versions below is how the fitted models are *reported*.

```

modelA <- lm(N_trips ~ Nr_People_H, data = data)
modelB <- lm(N_trips ~ Nr_People_H + Driv_Lic, data = data)
modelC <- lm(N_trips ~ Learners_students + Workers_Fulltime + Driv_Lic, data = data)
modelD <- vif_full_model
modelE <- lm(N_trips ~ Nr_People_H + Learners_students + Driv_Lic, data = data)
modelF <- lm(N_trips ~ Nr_People_H + Workers_Fulltime + Learners_students, data = data)

models <- list(A = modelA, B = modelB, C = modelC,
               D = modelD, E = modelE, F = modelF)

```

5.1 Version A - Base R: summary() printouts

Long lines from `summary()` (e.g. the `Call:`) are manually wrapped at 78 characters before being placed in a fenced block, so nothing runs off the page.

Model A : `N_trips ~ Nr_People_H`

Call:

```
lm(formula = N_trips ~ Nr_People_H, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.1149	-0.2996	-0.0403	0.1817	4.1817

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.780985	0.019963	39.12	<2e-16 ***
Nr_People_H	0.259329	0.007157	36.24	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6959 on 5380 degrees of freedom
(5 observations deleted due to missingness)

Multiple R-squared: 0.1962, Adjusted R-squared: 0.196

F-statistic: 1313 on 1 and 5380 DF, p-value: < 2.2e-16

Model B : $N_trips \sim Nr_People_H + Driv_Lic$

Call:

```
lm(formula = N_trips ~ Nr_People_H + Driv_Lic, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.0341	-0.3621	-0.1043	0.2475	4.2549

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.713830	0.021370	33.40	<2e-16 ***
Nr_People_H	0.257808	0.007113	36.25	<2e-16 ***
Driv_Lic	0.132613	0.015713	8.44	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6914 on 5379 degrees of freedom

(5 observations deleted due to missingness)

Multiple R-squared: 0.2067, Adjusted R-squared: 0.2064

F-statistic: 700.7 on 2 and 5379 DF, p-value: < 2.2e-16

Model C : $N_trips \sim \text{Learners_students} + \text{Workers_Fulltime} + \text{Driv_Lic}$

Call:

```
lm(formula = N_trips ~ Learners_students + Workers_Fulltime +  
    Driv_Lic, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.5201	-0.3902	-0.1098	0.2392	5.8902

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.10984	0.01214	91.428	< 2e-16 ***
Learners_students	1.20513	0.02110	57.122	< 2e-16 ***
Workers_Fulltime	0.28037	0.01475	19.013	< 2e-16 ***
Driv_Lic	0.09025	0.01481	6.095	1.17e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6022 on 5383 degrees of freedom

Multiple R-squared: 0.4141, Adjusted R-squared: 0.4138

F-statistic: 1268 on 3 and 5383 DF, p-value: < 2.2e-16

Model D : $N_trips \sim Nr_People_H + Elderly + Educ_HS + Educ_Tertiary + Workers_Fulltime + Workers_Parttime + Learners_students + Driv_Lic$

Call:

```
lm(formula = full_formula, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.4694	-0.3225	-0.0975	0.1641	4.3515

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.706093	0.016793	42.048	<2e-16 ***
Nr_People_H	0.129836	0.006671	19.464	<2e-16 ***
Elderly	-0.021507	0.008972	-2.397	0.0166 *
Educ_HS	0.233347	0.012779	18.260	<2e-16 ***
Educ_Tertiary	0.187435	0.018083	10.365	<2e-16 ***
Workers_Fulltime	0.144875	0.014090	10.282	<2e-16 ***
Workers_Parttime	0.336228	0.024048	13.982	<2e-16 ***
Learners_students	0.984266	0.020920	47.049	<2e-16 ***
Driv_Lic	0.040696	0.013166	3.091	0.0020 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4992 on 5307 degrees of freedom

(71 observations deleted due to missingness)

Multiple R-squared: 0.4781, Adjusted R-squared: 0.4773

F-statistic: 607.8 on 8 and 5307 DF, p-value: < 2.2e-16

Model E : $N_trips \sim Nr_People_H + Learners_students + Driv_Lic$

Call:

```
lm(formula = N_trips ~ Nr_People_H + Learners_students + Driv_Lic,  
    data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8759	-0.3244	-0.1441	0.1873	4.3676

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.818586	0.017927	45.66	<2e-16 ***
Nr_People_H	0.162760	0.006236	26.10	<2e-16 ***
Learners_students	1.040378	0.021337	48.76	<2e-16 ***
Driv_Lic	0.180322	0.013123	13.74	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5758 on 5378 degrees of freedom

(5 observations deleted due to missingness)

Multiple R-squared: 0.4499, Adjusted R-squared: 0.4496

F-statistic: 1466 on 3 and 5378 DF, p-value: < 2.2e-16

Model F : $N_trips \sim Nr_People_H + Workers_Fulltime + Learners_students$

Call:

```
lm(formula = N_trips ~ Nr_People_H + Workers_Fulltime + Learners_students,
    data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.7753	-0.3801	-0.1084	0.1744	4.4461

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.811424	0.016976	47.80	<2e-16 ***
Nr_People_H	0.148495	0.006153	24.14	<2e-16 ***
Workers_Fulltime	0.271651	0.012889	21.08	<2e-16 ***
Learners_students	1.036471	0.020824	49.77	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.563 on 5378 degrees of freedom

(5 observations deleted due to missingness)

Multiple R-squared: 0.474, Adjusted R-squared: 0.4737

F-statistic: 1615 on 3 and 5378 DF, p-value: < 2.2e-16

5.2 Version B - Package (broom): tidy coefficient and fit tables

```
ensure_pkg("broom")

for (nm in names(models)) {
  tidy_tbl <- broom::tidy(models[[nm]])
  needspace(nrow(tidy_tbl) + 10)
  cat("**Model", nm, ":", deparse(formula(models[[nm]])), "\n\n")
  tidy_tbl[, -1] <- round(tidy_tbl[, -1], 4)
  print(kable_safe(tidy_tbl, row.names = FALSE))
  cat("\n")
  glance_tbl <- broom::glance(models[[nm]])[c("r.squared", "adj.r.squared",
                                              "statistic", "p.value", "nobs")]
  glance_tbl <- round(glance_tbl, 4)
  colnames(glance_tbl) <- c("$R^2$", "$R^2_{adj}$", "statistic", "p.value", "nobs")
  print(kable_safe(glance_tbl, row.names = FALSE, escape = FALSE))
  cat("\n\n")
}
```

Model A : N_trips ~ Nr_People_H

term	estimate	std.error	statistic	p.value
(Intercept)	0.7810	0.0200	39.1219	0
Nr_People_H	0.2593	0.0072	36.2368	0

R^2	R^2_{adj}	statistic	p.value	nobs
0.1962	0.196	1313.108	0	5382

Model B : N_trips ~ Nr_People_H + Driv_Lic

term	estimate	std.error	statistic	p.value
(Intercept)	0.7138	0.0214	33.4026	0
Nr_People_H	0.2578	0.0071	36.2469	0
Driv_Lic	0.1326	0.0157	8.4395	0

R^2	R^2_{adj}	statistic	p.value	nobs
0.2067	0.2064	700.7369	0	5382

Model C : N_trips ~ Learners_students + Workers_Fulltime + Driv_Lic

term	estimate	std.error	statistic	p.value
(Intercept)	1.1098	0.0121	91.4284	0
Learners_students	1.2051	0.0211	57.1224	0
Workers_Fulltime	0.2804	0.0147	19.0127	0
Driv_Lic	0.0902	0.0148	6.0951	0

R^2	R^2_{adj}	statistic	p.value	nobs
0.4141	0.4138	1268.247	0	5387

Model D : N_trips ~ Nr_People_H + Elderly + Educ_HS + Educ_Tertiary + Workers_Fulltime + Workers_Parttime + Learners_students + Driv_Lic

term	estimate	std.error	statistic	p.value
(Intercept)	0.7061	0.0168	42.0480	0.0000
Nr_People_H	0.1298	0.0067	19.4641	0.0000
Elderly	-0.0215	0.0090	-2.3972	0.0166
Educ_HS	0.2333	0.0128	18.2600	0.0000
Educ_Tertiary	0.1874	0.0181	10.3653	0.0000
Workers_Fulltime	0.1449	0.0141	10.2822	0.0000
Workers_Parttime	0.3362	0.0240	13.9815	0.0000
Learners_students	0.9843	0.0209	47.0492	0.0000
Driv_Lic	0.0407	0.0132	3.0911	0.0020

R^2	R^2_{adj}	statistic	p.value	nobs
0.4781	0.4773	607.7632	0	5316

Model E : $N_trips \sim Nr_People_H + Learners_students + Driv_Lic$

term	estimate	std.error	statistic	p.value
(Intercept)	0.8186	0.0179	45.6624	0
Nr_People_H	0.1628	0.0062	26.1003	0
Learners_students	1.0404	0.0213	48.7583	0
Driv_Lic	0.1803	0.0131	13.7411	0

R^2	R^2_{adj}	statistic	p.value	nobs
0.4499	0.4496	1465.998	0	5382

Model F : $N_trips \sim Nr_People_H + Workers_Fulltime + Learners_students$

term	estimate	std.error	statistic	p.value
(Intercept)	0.8114	0.0170	47.7995	0
Nr_People_H	0.1485	0.0062	24.1355	0
Workers_Fulltime	0.2717	0.0129	21.0755	0
Learners_students	1.0365	0.0208	49.7738	0

R^2	R^2_{adj}	statistic	p.value	nobs
0.474	0.4737	1615.479	0	5382

6 5. Model comparison and parsimony discussion

The full predictor list per model is long for Model D, so - to keep every table column comfortably inside the page margins - it is listed once here as plain text (which wraps like any other paragraph) rather than repeated inside a table cell.

- **A:** $N_trips \sim Nr_People_H$
- **B:** $N_trips \sim Nr_People_H + Driv_Lic$
- **C:** $N_trips \sim Learners_students + Workers_Fulltime + Driv_Lic$
- **D:** $N_trips \sim Nr_People_H + Elderly + Educ_HS + Educ_Tertiary + Workers_Fulltime + Workers_Parttime + Learners_students + Driv_Lic$
- **E:** $N_trips \sim Nr_People_H + Learners_students + Driv_Lic$
- **F:** $N_trips \sim Nr_People_H + Workers_Fulltime + Learners_students$

6.1 Version A - Base R comparison table

```
comparison <- data.frame(
  Model = names(models),
  Predictors = sapply(models, get_predictor_str),
  R2 = sapply(models, function(m) round(summary(m)$r.squared, 4)),
  Adj_R2 = sapply(models, function(m) round(summary(m)$adj.r.squared, 4)),
  n = sapply(models, function(m) nobs(m)),
```

Table 3: Base R model comparison table (see predictor list above)

Model	R^2	R^2_{adj}	n
A	0.1962	0.1960	5382
B	0.2067	0.2064	5382
C	0.4141	0.4138	5387
D	0.4781	0.4773	5316
E	0.4499	0.4496	5382
F	0.4740	0.4737	5382

```
stringsAsFactors = FALSE
)

write.csv(comparison, "model_comparison.csv", row.names = FALSE)

comparison_display <- comparison[, c("Model", "R2", "Adj_R2", "n")]
colnames(comparison_display) <- c("Model", "$R^2$", "$R^2_{adj}$", "n")

kable_safe(comparison_display, row.names = FALSE, escape = FALSE,
            caption = "Base R model comparison table (see predictor list above)")
```

6.2 Version B - Package (broom + dplyr) comparison table

```
ensure_pkg("dplyr")
ensure_pkg("purrr")

comparison_broom <- purrr::map_dfr(names(models), function(nm) {
  g <- broom::glance(models[[nm]])
  dplyr::tibble(Model = nm,
                Predictors = get_predictor_str(models[[nm]]),
                R2 = round(g$r.squared, 4),
                Adj_R2 = round(g$adj.r.squared, 4),
                n = g$nobs)
})

comparison_broom_display <- comparison_broom[, c("Model", "R2", "Adj_R2", "n")]
colnames(comparison_broom_display) <- c("Model", "$R^2$", "$R^2_{adj}$", "n")

kable_safe(comparison_broom_display, row.names = FALSE, escape = FALSE,
            caption = "broom/dplyr model comparison table (see predictor list above)")
```

The two comparison tables agree exactly (checked above), as expected since both simply reformat the same fitted models.

Table 4: broom/dplyr model comparison table (see predictor list above)

Model	R^2	R^2_{adj}	n
A	0.1962	0.1960	5382
B	0.2067	0.2064	5382
C	0.4141	0.4138	5387
D	0.4781	0.4773	5316
E	0.4499	0.4496	5382
F	0.4740	0.4737	5382

6.2.1 Coefficient significance

```
sig_flags <- function(m) {
  cf <- summary(m)$coefficients
  cf <- cf[rownames(cf) != "(Intercept)", , drop = FALSE]
  data.frame(term = rownames(cf),
             t = round(cf[, "t value"], 2),
             p = round(cf[, "Pr(>|t|)"], 4),
             sig = cf[, "Pr(>|t|)" ] < 0.05)
}

for (nm in names(models)) {
  flags <- sig_flags(models[[nm]])
  needspace(nrow(flags) + 8)
  cat("**Model", nm, "**\n\n")
  print(kable_safe(flags, row.names = FALSE))
  cat("\n\n")
}
```

Model A

term	t	p	sig
Nr_People_H	36.24	0	TRUE

Model B

term	t	p	sig
Nr_People_H	36.25	0	TRUE
Driv_Lic	8.44	0	TRUE

Model C

term	t	p	sig
Learners_students	57.12	0	TRUE
Workers_Fulltime	19.01	0	TRUE
Driv_Lic	6.10	0	TRUE

Model D

term	t	p	sig
Nr_People_H	19.46	0.0000	TRUE
Elderly	-2.40	0.0166	TRUE
Educ_HS	18.26	0.0000	TRUE
Educ_Tertiary	10.37	0.0000	TRUE
Workers_Fulltime	10.28	0.0000	TRUE
Workers_Parttime	13.98	0.0000	TRUE
Learners_students	47.05	0.0000	TRUE
Driv_Lic	3.09	0.0020	TRUE

Model E

term	t	p	sig
Nr_People_H	26.10	0	TRUE
Learners_students	48.76	0	TRUE
Driv_Lic	13.74	0	TRUE

Model F

term	t	p	sig
Nr_People_H	24.14	0	TRUE
Workers_Fulltime	21.08	0	TRUE
Learners_students	49.77	0	TRUE

6.2.2 Parsimony commentary

The commentary below is computed directly from the fitted results above (not hard-coded), so it stays accurate if the underlying data changes.

Model A (single predictor Nr_People_H) already explains $R^2 = 0.196$ ($R^2_{adj} = 0.196$).

Among the reduced (non-D) models, Model F achieves the highest R^2_{adj} (0.474) using only 3 predictor(s): Nr_People_H + Workers_Fulltime + Learners_students.

Model D (all 8 predictors) reaches $R^2_{adj} = 0.477$ – a gain of just 0.0036 over Model F, for 5 extra predictors.

Manual VIFs for all 8 predictors are below 2 (max is 1.612), so multicollinearity is not severe enough on its own to destabilise Model D’s coefficients – all 8 coefficients in Model D remain statistically significant at the 5% level. However, several of the predictors are moderately correlated with each other (e.g. Nr_People_H-Elderly, Educ_Tertiary-Driv_Lic, Educ_Tertiary-Workers_Fulltime), so part of what they add in Model D duplicates explanatory power already captured by the core predictors (Nr_People_H, Learners_students, Workers_Fulltime, Driv_Lic).

Given that Model F recovers 99.2% of Model D’s R^2_{adj} with 5 fewer predictors, the marginal explanatory gain from fitting the full 8-predictor Model D does not appear to be worth its loss of parsimony: a small, well-chosen predictor set (household size, learners/students, workers, driver’s licence holding) captures almost all of the explanatory power at much lower risk of overfitting and unnecessary collinearity.

7 Outputs

Outputs written to the working directory:

- `figures/correlation_heatmap_baseR.png`
- `figures/correlation_ellipse.png`
- `figures/scatterplots_baseR.png`
- `figures/scatterplots_ggplot.png`
- `model_comparison.csv`